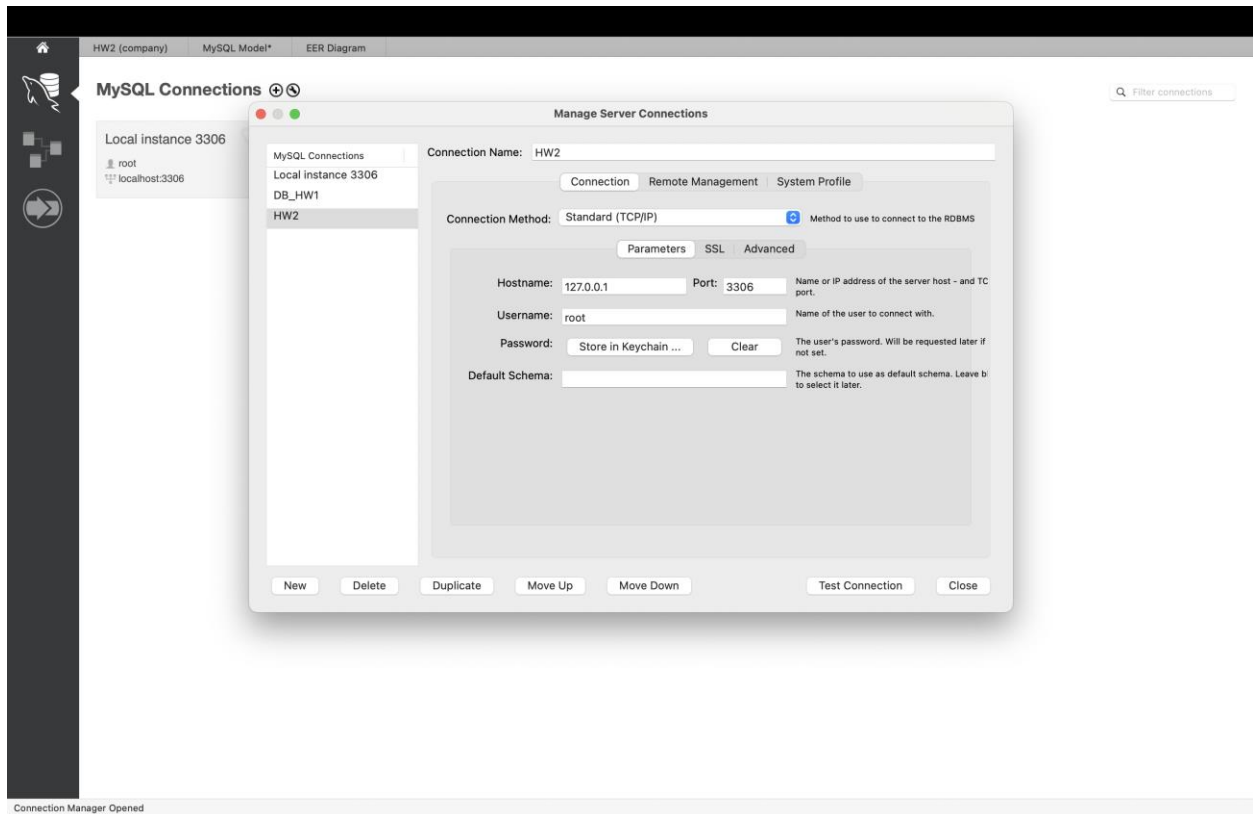


Jeevan Kumar Banoth – 02105145

CIS 552: Database Design

Homework – 2:

➤ Creation of a connection with Database: -



As you can see above First, I made the connection with the SQL database by creating a new connection called HW2 and tested the connection is successful or not.

After that I opened that connection and started writing the SQL code for the given problem.

1. Building an ER Schema Using SQL developer Data model.

Started by creating the database named “COMPANY” and opened that database by using command in the next line.

Then, I created the tables and defined the columns, constraints as per fig 6.1 in chapter 6, this usually includes defining primary keys, foreign keys, and other constraints.

HW2 (company)

Administration Schemas HW2*

SCHEMAS

Filter objects

COMPANY

- Tables
- Views
- Stored Procedures
- Functions
- sys
- UniversityDB

```
1 CREATE DATABASE COMPANY;
2
3 USE COMPANY;
4
5 CREATE TABLE EMPLOYEE (
6     First_Name VARCHAR(15) NOT NULL,
7     Middle_Init CHAR,
8     Last_Name VARCHAR(30) NOT NULL,
9     Social_Security_Number CHAR(9) NOT NULL,
10    Birth_Date DATE NOT NULL,
11    Home_Address VARCHAR(30),
12    Gender CHAR,
13    Salary DECIMAL(10, 2),
14    Supervisor_SSN CHAR(9),
15    Department_Number INT NOT NULL,
16    PRIMARY KEY (Social_Security_Number)
17 );
18
19 CREATE TABLE DEPARTMENT(
20     Department_Name VARCHAR(15) NOT NULL,
```

Result Grid

Filter Rows: Search Export

First_Name Last_Name

Result 4 Result 5 Result 6 Read Only

Action Output

	Time	Action	Response	Duration / Fetch Time
1	10:35:51	CREATE DATABASE COMPANY	Error Code: 1007. Can't create database 'COMPANY';...	0.00033 sec
2	10:35:51	USE COMPANY	0 row(s) affected	0.00011 sec
3	10:35:51	CREATE TABLE DEPARTMENT (DNUMBER INT PRIMARY KEY, DNAME VARCHAR(50) NOT NULL, PRIMARY KEY (DNUMBER))	0 row(s) affected	0.010 sec
4	10:35:51	CREATE TABLE EMPLOYEE (SSN VARCHAR(9) PRIMARY KEY, FNAME VARCHAR(50) NOT NULL, MNAME VARCHAR(15) NOT NULL, LNAME VARCHAR(30) NOT NULL, BDATE DATE NOT NULL, HADDR VARCHAR(30), GENDER CHAR, SALARY DECIMAL(10, 2), SUPERVISOR_SSN CHAR(9), DEPT_NUM INT NOT NULL, PRIMARY KEY (SSN))	0 row(s) affected	0.0049 sec
5	10:35:51	CREATE TABLE PROJECT (PNUMBER INT PRIMARY KEY, PNAME VARCHAR(50) NOT NULL, PLOC VARCHAR(15) NOT NULL, DEPT_NUM INT NOT NULL, PRIMARY KEY (PNUMBER))	0 row(s) affected	0.0046 sec
6	10:35:51	CREATE TABLE WORKS_ON (ESSN VARCHAR(9), PNO INT, HOURS DECIMAL(3, 2), PRIMARY KEY (ESSN, PNO))	0 row(s) affected	0.0028 sec
7	10:35:51	CREATE TABLE DEPENDENT (ESSN VARCHAR(9), DEPENDENT_NAME VARCHAR(30), PRIMARY KEY (ESSN, DEPENDENT_NAME))	0 row(s) affected	0.0020 sec

Query Completed



HW2 (company)

Administration Schemas HW2*

SCHEMAS

Filter objects

COMPANY

- Tables
- Views
- Stored Procedures
- Functions
- sys
- UniversityDB

```
23 PRIMARY KEY (Department_Number),
24 UNIQUE (Manager_SSN),
25 FOREIGN KEY (Manager_SSN) REFERENCES EMPLOYEE(Social_Security_Number)
26 );
27
28
29 CREATE TABLE DEPARTMENT_LOCATIONS (
30     Department_Number INT NOT NULL,
31     Location VARCHAR(15),
32     PRIMARY KEY (Department_Number, Location),
33     FOREIGN KEY (Department_Number) REFERENCES DEPARTMENT(Department_Number)
34 );
35
36 CREATE TABLE PROJECT (
37     Project_Name VARCHAR(15) NOT NULL,
38     Project_Number INT NOT NULL,
39     Project_Location VARCHAR(15),
40     Department_Num INT NOT NULL,
41     PRIMARY KEY (Project_Number),
42     UNIQUE (Project_Name),
43     FOREIGN KEY (Department_Num) REFERENCES DEPARTMENT(Department_Number)
44 );
```

Result Grid

Filter Rows: Search Export

First_Name Last_Name

Result 4 Result 5 Result 6 Read Only

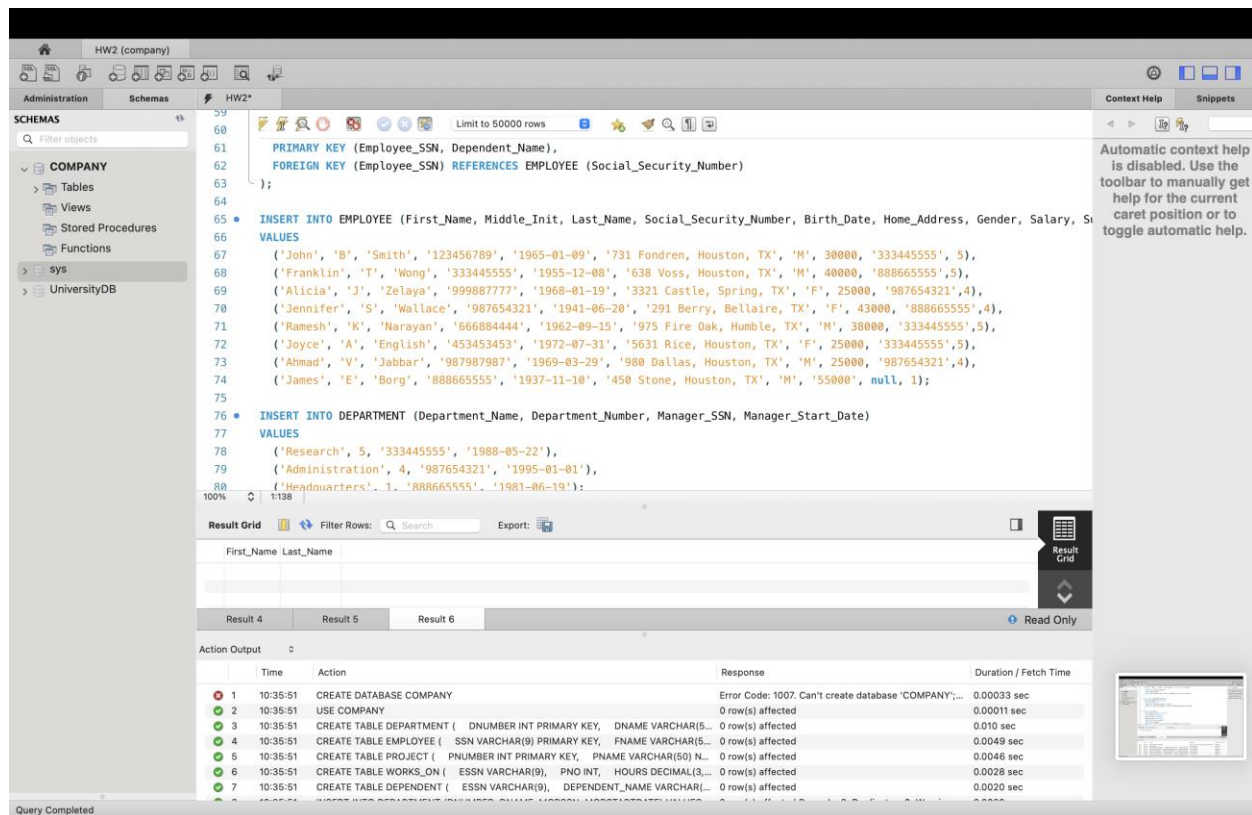
Action Output

	Time	Action	Response	Duration / Fetch Time
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2	10:35:51	USE COMPANY	0 row(s) affected	0.00011 sec
3	10:35:51	CREATE TABLE DEPARTMENT (DNUMBER INT PRIMARY KEY, DNAME VARCHAR(50) NOT NULL, PRIMARY KEY (DNUMBER))	0 row(s) affected	0.010 sec
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6	10:35:51	CREATE TABLE WORKS_ON (ESSN VARCHAR(9), PNO INT, HOURS DECIMAL(3, 2), PRIMARY KEY (ESSN, PNO))	0 row(s) affected	0.0028 sec
7	10:35:51	CREATE TABLE DEPENDENT (ESSN VARCHAR(9), DEPENDENT_NAME VARCHAR(30), PRIMARY KEY (ESSN, DEPENDENT_NAME))	0 row(s) affected	0.0020 sec

Query Completed

Next, used “INSERT” command or MySQL workbench’s GUI to populate the data as per the figure instructions given in the question.

After the creation of database and table definitions was done, I went through data insertion step, in this one we will be going to enter all the data that we have for like SSN (Social Security Number), employee name, gender, birth date, salary etc...



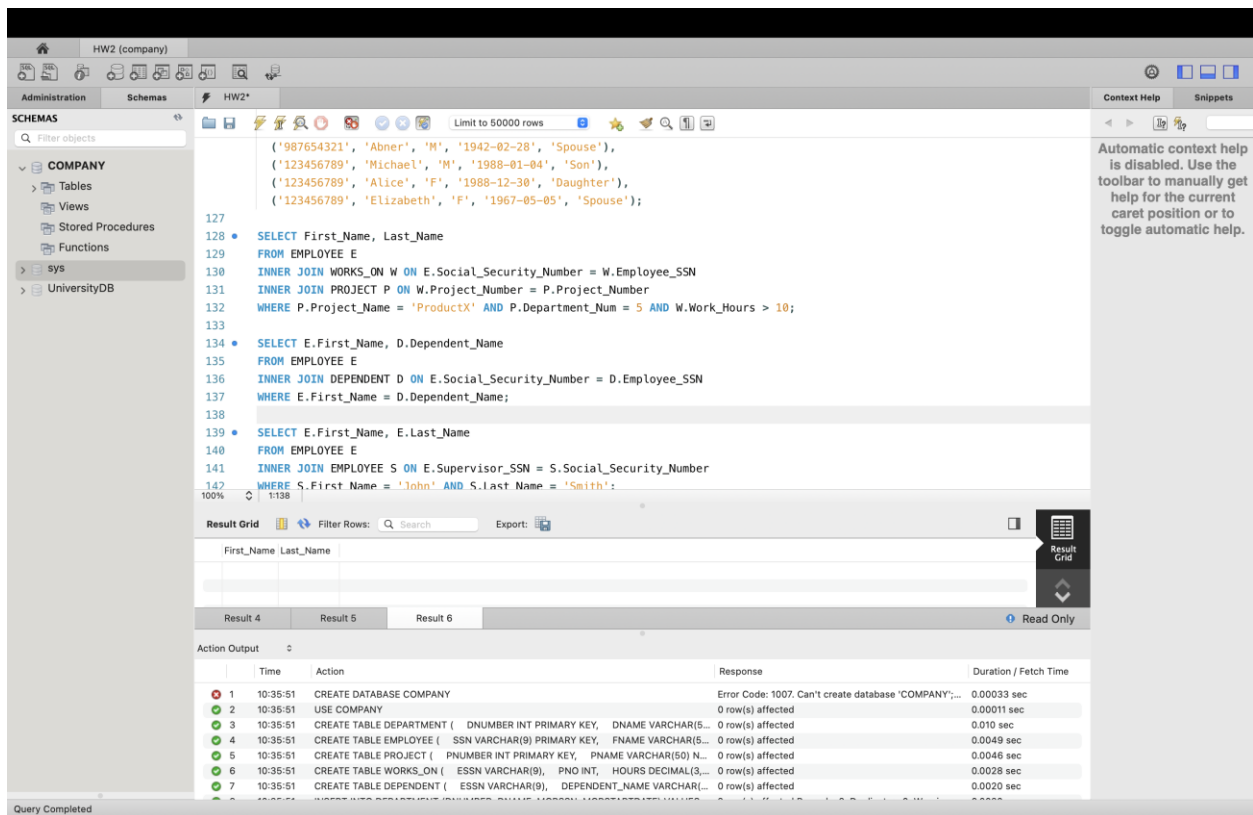
As shown above, I have inserted all the details of it and as well for the department section also I gave the Department name, number, SSN etc...

And when it comes to the project part, I gave the project details like name of it, location, dept number.

And in the working hours section I gave the project number, Hours and employees SSN number, same way I gave the dependent details too like name, and which employee is that associated with and their SSN number.

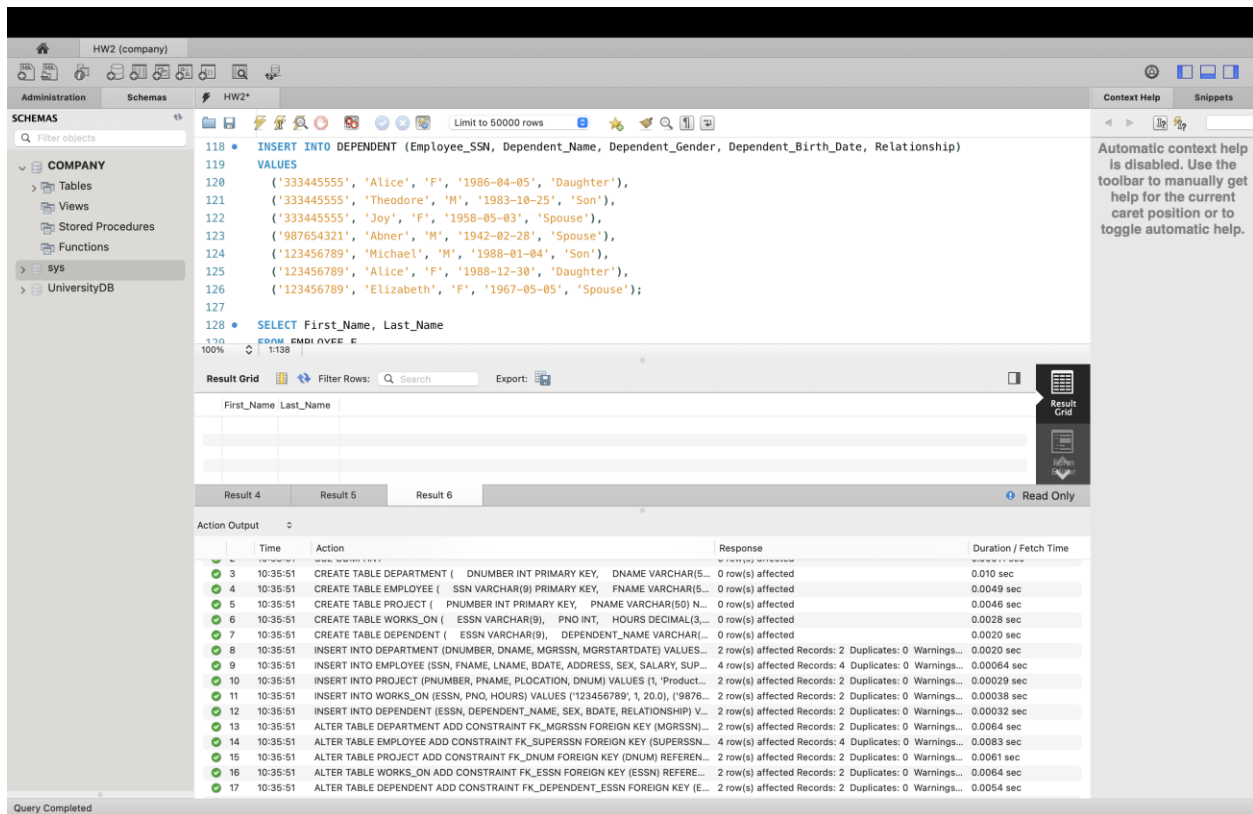
Queries – Data extraction

- Obtaining the identities of all Department 5 workers who devote more than 10 hours a week to the Project
- List the names of all employees who have a dependent with the same first name as themselves.
- Find the names of all employees who are directly supervised by ‘John Smith’:

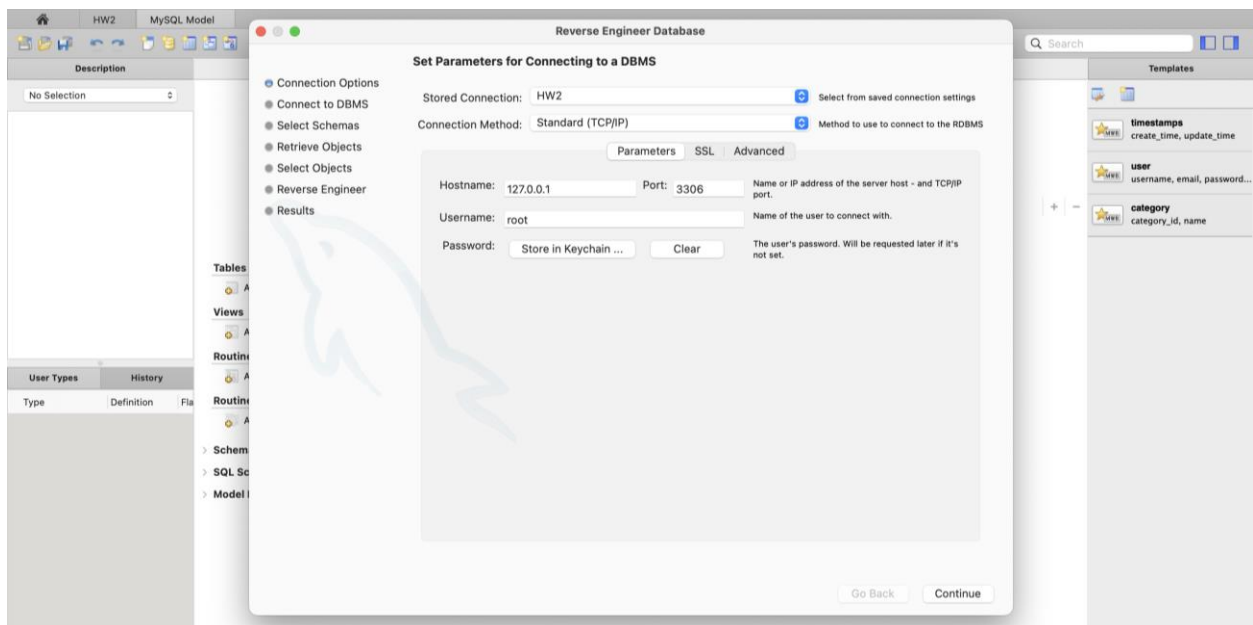


The above shown are the queries that I wrote for this one with the requirements.

The below attached Image shows the result and execution of the code is done successfully.



After this code has been run successfully, by using reverse engineering we will plot the diagram in SQL workbench. The plot and SQL code of this one had been attached to you.



Conclusion: -

In this scholarly work, the "COMPANY" database was set up using Microsoft SQL Server Management Studio.

It comprised of such related tables as EMPLOYEE, DEPARTMENT, DEPT_LOCATIONS, PROJECT, WORKS_ON, and DEPENDENT, all of them having dissimilar characteristics and limitations.

I also inserted data into these tables to show how results come out during performance of three different queries.

I designed these queries to search for certain information in a database like recognizing staff members belonging to department 5 that spend more than ten hours a week on the "Project X" scheme, identifying members of the personnel having relatives with similar first names, and finding all employees reporting their activities solely to "John Smith."